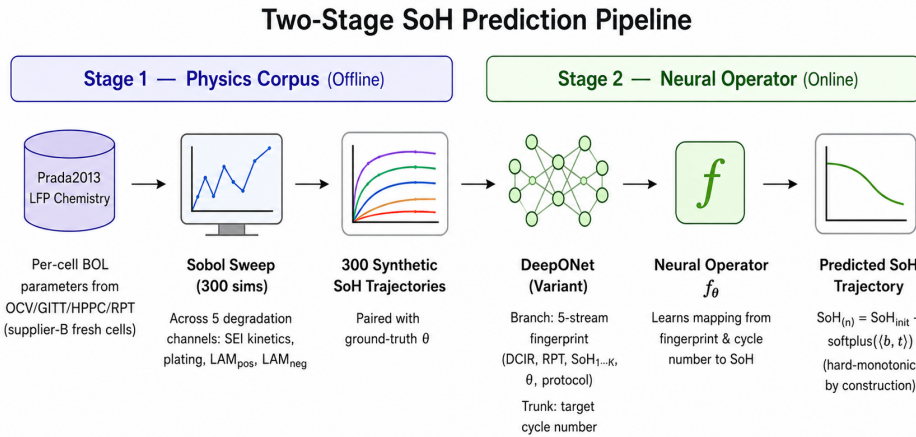


# From a single characterisation snapshot to end-of-life: a PyBaMM-trained neural operator for second-life LFP RUL

Krishna Chaitanya Vaddepally  
Turno (Blubble Pvt. Ltd.), Bengaluru, India  
krishna.vaddepally@turno.club

Second-life reuse of used lithium-iron-phosphate (LFP) cells for stationary energy storage is bottlenecked by characterisation cost. Per-cell PyBaMM [1] calibration of degradation parameters against measured state-of-health (SoH) trajectories requires at least 400 cycles per cell [2] to reliably extrapolate to end-of-life; below that, the fitted rates become under-determined. Prior second-life work [3] and universal-differential-equation degradation [4] improve the quality of the per-cell fit but do not shrink the cycling budget itself. Reducing that budget requires transferring knowledge from a synthetic parameter-space corpus to individual cells rather than fitting each from scratch.

The approach has two stages: an offline physics corpus built with PyBaMM's DFN model on the Prada2013 [2] LFP chemistry and per-cell beginning-of-life parameters identified from OCV, GITT, HPPC and RPT tests on a fresh supplier-B cohort; and an online neural operator — a DeepONet [5] variant conditioned on the identified degradation vector  $\theta$  — that maps a five-stream characterisation fingerprint to the SoH trajectory at any queried cycle. Figure 1 summarises the flow.



**Figure 1.** Two-stage pipeline: a 300-trajectory PyBaMM Sobol corpus (five degradation channels, 25 °C) trains a  $\theta$ -conditioned DeepONet that predicts SoH from a single characterisation snapshot.

End-to-end validation on a single supplier-B cell drives both the BOL identification and the degradation-parameter fit from that cell's own data. Fitting the four SEI and plating parameters against its 150-cycle measured Longterm record yields **0.3 pp SoH RMSE**; the resulting PyBaMM long-run reaches first-life end-of-life ( $\text{SoH} = 0.80$ ) at **cycle 3921** with a smooth monotonic fade, consistent with major LFP manufacturer datasheet cycle-life claims at 0.5 C. In this presentation the trained operator's predictions on 20 held-out cells across three commercial LFP suppliers will be shown, together with the identifiability limits of parameter fitting from short cycling records and extensions to non-isothermal deployment.

## References

1. V. Sulzer *et al.*, PyBaMM, *J. Open Res. Softw.* **9**(1), 14–21 (2021).
2. M. Prada *et al.*, Simplified electrochemical and thermal model of  $\text{LiFePO}_4$ -graphite Li-ion batteries, *J. Electrochem. Soc.* **160**(4), A616–A628 (2013).
3. O. Ouledboutaarija *et al.*, Advancing Second-Life Lithium-Ion Batteries, PyBaMM Conf. 2025, p. 19.
4. J. A. Kuzhiyil *et al.*, Enhancing Generalisability of Physics-based Battery Degradation Using UDE, PyBaMM Conf. 2025, p. 12.
5. L. Lu *et al.*, DeepONet: learning nonlinear operators for identifying differential equations, *Nat. Mach. Intell.* **3**, 218–229 (2021).